

CNS INFECTIONS

Introduction

- Include bacterial meningitis, viral meningitis, encephalitis and focal infections such as brain abscess.
- ✓ Key goals of early management are to emergently distinguish between these conditions, identify the responsible pathogen, and initiate appropriate antimicrobial therapy.

Acute Bacterial Meningitis

- an acute purulent infection within the subarachnoid space.
- associated with a CNS inflammatory reaction that may result in decreased consciousness, seizures, raised intracranial pressure (ICP), and stroke.
- The meninges, the subarachnoid space, and the brain parenchyma are all frequently involved in the inflammatory reaction (*meningoencephalitis*).

Epidemiology

- Bacterial meningitis is the most common form of suppurative CNS infection,
- the organisms most commonly responsible for community-acquired bacterial meningitis are *Streptococcus pneumoniae* , *N.meningitidis* , *H. influenzae* , group B streptococci, and *Listeria monocytogenes* .
- *N. meningitidis* is the causative organism of recurring epidemics of meningitis every 8 to 12 years.

Etiologies

S. pneumoniae

- is the most common cause of meningitis in adults >20 years of age
- There are a number of predisposing conditions that increase the risk of pneumococcal meningitis, the most important of which is **pneumococcal pneumonia**.
- ✓ Additional risk factors include coexisting acute or chronic pneumococcal **sinusitis or otitis media, alcoholism, diabetes, splenectomy, hypogammaglobulinemia, complement deficiency**, and head trauma with basilar skull fracture and CSF rhinorrhea.
- Mortality remains ~20% despite antibiotic therapy.

N. meningitidis

- accounts for 25% of all cases of bacterial meningitis and for up to 60% of cases in children and young adults between the ages of 2 and 20.
- The presence of **petechial or purpuric skin lesions** can provide an important clue to the diagnosis of meningococcal infection.
- In some patients the disease is fulminant, progressing to death within hours of symptom onset.
- Infection may be initiated by nasopharyngeal colonization, which can result in either an asymptomatic carrier state or invasive meningococcal disease.

- The risk of invasive disease following nasopharyngeal colonization depends on **both bacterial virulence factors and host immune defense mechanisms**, including the host's capacity to produce antimeningococcal antibodies and to lyse meningococci by both classic and alternative complement pathways.
- Individuals with deficiencies of any of the complement components are highly susceptible to meningococcal infections.

- ✓ More frequently, *H. influenzae* causes meningitis in **unvaccinated children and adults.**
- *Staphylococcus aureus* and coagulase-negative staphylococci are important causes of meningitis that occurs following invasive neurosurgical procedures.

Pathophysiology

- initially colonize the nasopharynx by attaching to epithelial cells.
- Bacteria are transported across epithelial cells in membrane
- bound vacuoles to the intravascular space or invade the intravascular space by creating separations in the apical tight junctions of columnar epithelial cells.

- ✓ Once in the bloodstream, bacteria are able to avoid phagocytosis by neutrophils and classic complement-mediated bactericidal activity because of the presence of a polysaccharide capsule.
- ✓ Blood born bacteria can reach the intraventricular choroid plexus, directly infect choroid plexus epithelial cells, and gain access to the CSF.

- Normal CSF contains few white blood cells (WBCs) and relatively small amounts of complement proteins and immunoglobulins preventing effective opsonization of bacteria.
- ✓ Phagocytosis of bacteria is further impaired by the fluid nature of CSF, which is less conducive to phagocytosis than a solid tissue substrate.

Clinical Presentation

- Meningitis can present as either
 - acute fulminant illness that progresses rapidly in a few hours
 - subacute infection that progressively worsens over several days.
- The classic clinical triad of meningitis **is fever, headache, and nuchal rigidity.**
- A decreased level of consciousness occurs in >75% of patients and can vary from lethargy to coma.
- Nausea, vomiting, and photophobia are also common complaints.

- Seizures occur as part of the initial presentation of bacterial meningitis or during the course of the illness in 20–40% of patients.
- ✓ Focal seizures are usually due to focal arterial ischemia or infarction, cortical venous thrombosis with hemorrhage, or focal edema.
- ✓ Generalized seizure activity and status epilepticus may be due to hyponatremia or cerebral anoxia

- **Raised ICP** is an expected complication of bacterial meningitis and the major cause of obtundation and coma in this disease.
- ✓ Signs of increased ICP include a deteriorating or **reduced level of consciousness, papilledema, dilated poorly reactive pupils, sixth nerve palsies, decerebrate posturing, and the Cushing reflex (bradycardia, hypertension, and irregular respirations).**
- ✓ The most disastrous complication of increased ICP is **cerebral herniation.**
- ✓ The incidence of herniation in patients with bacterial meningitis has been reported to occur in as few as 1% to as many as 8% of cases.

Diagnosis

- Blood cultures should be immediately obtained and empirical antimicrobial therapy initiated without delay
- The diagnosis of bacterial meningitis is made by examination of the CSF.
- ✓ The need to obtain neuroimaging studies (CT or MRI) prior to LP requires clinical judgment.
- ✓ In an immunocompetent patient with no known history of recent head trauma, a normal level of consciousness, and no evidence of papilledema or focal neurologic deficits, it is considered safe to perform LP without prior neuroimaging studies.

Diagnosis continued..

- ✓ If LP is delayed in order to obtain neuroimaging studies, empirical antibiotic therapy should be initiated after blood cultures are obtained.
- ✓ Antibiotic therapy initiated a few hours prior to LP will not significantly alter the CSF WBC count or glucose concentration, nor is it likely to prevent visualization of organisms by Gram's stain or detection of bacterial nucleic acid by polymerase chain reaction (PCR) assay.

Table 376-2 Cerebrospinal Fluid (CSF) Abnormalities in Bacterial Meningitis

Opening pressure	>180 mmH ₂ O
White blood cells	10/ μ L to 10,000/ μ L; neutrophils predominate
Red blood cells	Absent in nontraumatic tap
Glucose	<2.2 mmol/L (<40 mg/dL)
CSF/serum glucose	<0.4
Protein	>0.45 g/L (>45 mg/dL)
Gram's stain	Positive in >60%
Culture	Positive in >80%
Latex agglutination	May be positive in patients with meningitis due to <i>S. pneumoniae</i> , <i>N. meningitidis</i> , <i>H. influenzae</i> type b, <i>E. coli</i> , group B streptococci
Limulus lysate	Positive in cases of gram-negative meningitis
PCR	Detects bacterial DNA

Acute Bacterial Meningitis:Treatment

- Empirical Antimicrobial Therapy
- ✓ Bacterial meningitis is a medical emergency.
- ✓ The goal is to begin antibiotic therapy within 60 min of a patient's arrival in the emergency room.
- Ceftriaxone or cefotaxime provide good coverage for susceptible *S. pneumoniae*, group B streptococci, and *H. influenzae* and adequate coverage for *N. meningitidis*.

- Ampicillin should be added to the empirical regimen for coverage of *L. monocytogenes* in individuals <3 months of age, those >55, or those with suspected impaired cell-mediated immunity because of chronic illness, organ transplantation, pregnancy, malignancy, or immunosuppressive therapy

Specific Antimicrobial Therapy

Meningococcal Meningitis

- Although ceftriaxone / cefotaxime provide adequate empirical coverage for *N. meningitidis*, penicillin G remains the antibiotic of choice for meningococcal meningitis caused by susceptible strains.
- A 7-day course of intravenous antibiotic therapy is adequate for uncomplicated meningococcal meningitis.

Table 376-3 Antimicrobial Therapy of CNS Bacterial Infections Based on Pathogen^a

Organism	Antibiotic
<i>Neisseria meningitides</i>	
Penicillin-sensitive	Penicillin G or ampicillin
Penicillin-resistant	Ceftriaxone or cefotaxime
<i>Streptococcus pneumoniae</i>	
Penicillin-sensitive	Penicillin G
Penicillin-intermediate	Ceftriaxone or cefotaxime
Penicillin-resistant	(Ceftriaxone or cefotaxime) + vancomycin
Gram-negative bacilli (except <i>Pseudomonas</i> spp.)	Ceftriaxone or cefotaxime
<i>Pseudomonas aeruginosa</i>	Ceftazidime or cefepime or meropenem
<i>Staphylococci</i> spp.	
Methicillin-sensitive	Nafcillin
Methicillin-resistant	Vancomycin
<i>Listeria monocytogenes</i>	Ampicillin + gentamicin
<i>Haemophilus influenzae</i>	Ceftriaxone or cefotaxime
<i>Streptococcus agalactiae</i>	Penicillin G or ampicillin
<i>Bacteroides fragilis</i>	Metronidazole
<i>Fusobacterium</i> spp.	Metronidazole

Adjunct Therapy

- The release of bacterial cell-wall components by bactericidal antibiotics leads to the production of the inflammatory cytokines IL-1 and TNF in the subarachnoid space.
- **Dexamethasone** exerts its beneficial effect by inhibiting the synthesis of IL-1 and TNF at the level of mRNA, decreasing CSF outflow resistance, and stabilizing the blood-brain barrier.

Adjunctive Therapy..

- The rationale for giving dexamethasone 20 min before antibiotic therapy is that dexamethasone inhibits the production of TNF by macrophages and microglia only if it is administered before these cells are activated by endotoxin.
- Dexamethasone does not alter TNF production once it has been induced.
- The results of clinical trials of dexamethasone therapy in children, predominantly with meningitis due to *H. influenzae* and *S. pneumoniae*, have demonstrated its efficacy in decreasing meningeal inflammation and neurologic sequelae such as the incidence of sensorineural hearing loss.

Increased Intracranial Pressure

- Emergency treatment of increased ICP includes
 - ✓ elevation of the patient's head to 30° ,
 - ✓ intubation and hyperventilation (Pa_{CO_2} 25–30 mmHg), and
 - ✓ mannitol.
- Patients with increased ICP should be managed in an intensive care unit; accurate ICP measurements are best obtained with an ICP monitoring device.

Prognosis

- Mortality is 3–7% for meningitis caused by *H. influenzae*, *N. meningitidis*, or group B streptococci; 15% for that due to *L. monocytogenes*; and 20% for *S. pneumoniae*.
- In general, the risk of death from bacterial meningitis increases with
 - ✓ decreased level of consciousness on admission,
 - ✓ onset of seizures within 24 h of admission,
 - ✓ signs of increased ICP,
 - ✓ young age (infancy) and age >50,
 - ✓ the presence of comorbid conditions including shock and/or the need for mechanical ventilation

- ✓ delay in the initiation of treatment.
- ✓ Decreased CSF glucose concentration [<2.2 mmol/L (<40 mg/dL)] and markedly increased CSF protein concentration [>3 g/L (>300 mg/dL)] have been predictive of increased mortality and poorer outcomes in some series.
- Moderate or severe sequelae occur in $\sim 25\%$ of survivors, although the exact incidence varies with the infecting organism.
- ✓ Common sequelae include decreased intellectual function, memory impairment, seizures, hearing loss and dizziness, and gait disturbances

Brain Abscess

- A brain abscess is a focal, suppurative infection within the brain parenchyma, typically surrounded by a vascularized capsule.
- The term *cerebritis* is often employed to describe a nonencapsulated brain abscess.
- Predisposing conditions include **otitis media and mastoiditis, paranasal sinusitis, pyogenic infections in the chest or other body sites, penetrating head trauma or neurosurgical procedures, and dental infections.**

- In immunocompetent individuals the most important pathogens are *Streptococcus* spp. [anaerobic, aerobic, and viridans], Enterobacteriaceae [*Proteus* spp., *E. coli* sp., *Klebsiella* spp.], anaerobes [e.g., *Bacteroides* spp., *Fusobacterium* spp.], and staphylococci.
- In immunocompromised hosts with underlying HIV infection, organ transplantation, cancer, or immunosuppressive therapy, most brain abscesses are caused by *Nocardia* spp., *Toxoplasma gondii*, *Aspergillus* spp., *Candida* spp., and *C. neoformans*.
- mycobacterial infection (tuberculoma) can cause focal CNS mass lesions.

Pathophysiology

- A brain abscess may develop
 - by **direct spread** from a contiguous cranial site of infection, such as paranasal sinusitis, otitis media, mastoiditis, or dental infection;
 - following **head trauma or a neurosurgical procedure**; or
 - as a **result of hematogenous** spread from a remote site of infection.
 - In up to 25% of cases, no obvious primary source of infection is apparent (**cryptogenic brain abscess**).

- **Approximately one-third of brain abscesses** are associated with **otitis media and mastoiditis**.
- ✓ Otogenic abscesses occur predominantly in the temporal lobe (55–75%) and cerebellum (20–30%).
- ✓ Common organisms include streptococci, *Bacteroides* spp., *Pseudomonas* spp., *Haemophilus* spp., and Enterobacteriaceae.
- Abscesses that develop as a result of direct spread of infection from the frontal, ethmoidal, or sphenoidal sinuses and those that occur due to dental infections are usually located in **the frontal lobes**.

- Approximately 10% of brain abscesses are associated with paranasal sinusitis, and this association is particularly strong in young males in their second and third decades of life.
- ✓ The most common pathogens in brain abscesses associated with paranasal sinusitis are streptococci (especially *S. milleri*), *Haemophilus* spp., *Bacteroides* spp., *Pseudomonas* spp., and *S. aureus*.
- Dental infections are associated with ~2% of brain abscesses, although it is often suggested that many "cryptogenic" abscesses are in fact due to dental infections.
- ✓ The most common pathogens in this setting are streptococci, staphylococci, *Bacteroides* spp., and *Fusobacterium* spp

- Hematogenous abscesses account for $\sim 25\%$ of brain abscesses.
- ✓ Hematogenous abscesses are often multiple, and multiple abscesses often (50%) have a hematogenous origin.
- ✓ These abscesses show a predilection for the territory of the middle cerebral artery (i.e., posterior frontal or parietal lobes).
- ✓ The microbiology of hematogenous abscesses is dependent on the primary source of infection.

Pathogenesis and Histopathology

- The intact brain parenchyma is relatively resistant to infection.
- Results of experimental models of brain abscess formation suggest that for bacterial invasion of brain parenchyma to occur, there must be preexisting or concomitant areas of ischemia, necrosis, or hypoxia in brain tissue.
- Once bacteria have established infection, brain abscess frequently evolves through a series of stages, influenced by the nature of the infecting organism and by the immunocompetence of the host.

- The **early cerebritis stage (days 1–3)** is characterized by a perivascular infiltration of inflammatory cells, which surround a central core of coagulative necrosis. Marked edema surrounds the lesion at this stage.
- In the **late cerebritis stage (days 4–9)**, pus formation leads to enlargement of the necrotic center, which is surrounded at its border by an inflammatory infiltrate of macrophages and fibroblasts.
- A thin capsule of fibroblasts and reticular fibers gradually develops, and the surrounding area of cerebral edema becomes more distinct than in the previous stage.

- **The third stage**, early capsule formation (days 10–13), is characterized by the formation of a capsule that is better developed on the cortical than on the ventricular side of the lesion.
- **The final stage**, late capsule formation (day 14 and beyond), is defined by a well-formed necrotic center surrounded by a dense collagenous capsule.
- The surrounding area of cerebral edema has regressed, but marked gliosis with large numbers of reactive astrocytes has developed outside the capsule.
- This gliotic process may contribute to the development of seizures as a sequelae of brain abscess.

Clinical Presentation

- A brain abscess typically presents as an expanding intracranial mass lesion rather than as an infectious process.
- The classic clinical triad of **headache, fever, and a focal neurologic deficit** is present in <50% of cases.
- The most common symptom in patients with a brain abscess is headache, occurring in >75% of patients.
- The headache is often characterized as a constant, dull, aching sensation, either hemicranial or generalized, and it becomes progressively more severe and refractory to therapy.

- Fever is present in only 50% of patients at the time of diagnosis, and its absence should not exclude the diagnosis.
- The new onset of focal or generalized seizure activity is a presenting sign in 15–35% of patients.
- Focal neurologic deficits including hemiparesis, aphasia, or visual field defects are part of the initial presentation in >60% of patients

- The clinical presentation of a brain abscess depends on its location, the nature of the primary infection if present, and the level of the ICP.
- ✓ **Hemiparesis** is the most common localizing sign of a **frontal lobe abscess**.
- ✓ **A temporal lobe abscess** may present with a disturbance of **language (dysphasia)** or an **upper homonymous quadrantanopia**.
- ✓ **Nystagmus and ataxia** are signs of a cerebellar abscess.
- ✓ Signs of raised ICP—papilledema, nausea and vomiting, and drowsiness or confusion—can be the dominant presentation of some abscesses, particularly those in the cerebellum.

Diagnosis

- Diagnosis is made by neuroimaging studies.
- ✓ MRI is better than CT for demonstrating abscesses in the early (cerebritis) stages and is superior to CT for identifying abscesses in the posterior fossa.

- Microbiologic diagnosis of the etiologic agent is most accurately determined by Gram's stain and culture of abscess material obtained by stereotactic needle aspiration.
- ✓ Aerobic and anaerobic bacterial cultures and mycobacterial and fungal cultures should be obtained.
- ✓ Blood cultures are positive in $\sim 10\%$ of cases overall but may be positive in $>85\%$ of patients with abscesses due to *Listeria*.

- LP should not be performed in patients with known or suspected focal intracranial infections such as abscess or empyema; as CSF analysis contributes nothing to diagnosis or therapy, and LP increases the risk of herniation.
- Additional laboratory studies may provide clues to the diagnosis of brain abscess in patients with a CNS mass lesion.
- ✓ About 50% of patients have a peripheral leukocytosis, 60% an elevated ESR, and 80% an elevated C-reactive protein.

Brain Abscess: Treatment

- Optimal therapy of brain abscesses involves a combination of high-dose parenteral antibiotics and neurosurgical drainage.
- Empirical therapy of community-acquired brain abscess in an immunocompetent patient typically includes a third-generation cephalosporin (e.g., cefotaxime or ceftriaxone) and metronidazole .
- In patients with penetrating head trauma or recent neurosurgical procedures, treatment should include ceftazidime as the third-generation cephalosporin to enhance coverage of *Pseudomonas* spp. and vancomycin for coverage of staphylococci.

- Aspiration and drainage of the abscess under stereotactic guidance are beneficial for both diagnosis and therapy. Empirical antibiotic coverage should be modified based on the results of Gram's stain and culture of the abscess contents.
- Complete excision of a bacterial abscess via craniotomy or craniectomy is generally reserved for multiloculated abscesses or those in which stereotactic aspiration is unsuccessful.

- Medical therapy alone is not optimal for treatment of brain abscess and should be reserved for patients whose abscesses are neurosurgically inaccessible, for patients with small ($<2\text{--}3\text{ cm}$) or nonencapsulated abscesses (cerebritis), and patients whose condition is too tenuous to allow performance of a neurosurgical procedure.
- All patients should receive a minimum of 6–8 weeks of parenteral antibiotic therapy.
- In addition to surgical drainage and antibiotic therapy, patients should receive prophylactic anticonvulsant therapy because of the high risk ($\sim 35\%$) of focal or generalized seizures.

- Anticonvulsant therapy is continued for at least 3 months after resolution of the abscess, and decisions regarding withdrawal are then based on the EEG.
- If the EEG is abnormal, anticonvulsant therapy should be continued.
- If the EEG is normal, anticonvulsant therapy can be slowly withdrawn, with close follow-up and repeat EEG after the medication has been discontinued

- Glucocorticoids should not be given routinely to patients with brain abscesses.
- ✓ Intravenous dexamethasone therapy (10 mg every 6 h) is usually reserved for patients with substantial periabscess edema and associated mass effect and increased ICP.
- ✓ Dexamethasone should be tapered as rapidly as possible to avoid delaying the natural process of encapsulation of the abscess.
- Serial MRI or CT scans should be obtained on a monthly or twice-monthly basis to document resolution of the abscess.
- ✓ More frequent studies (e.g., weekly) are probably warranted in the subset of patients who are receiving antibiotic therapy alone.

Prognosis

- The mortality of brain abscess has declined in parallel with the development of enhanced neuroimaging techniques, improved neurosurgical procedures for stereotactic aspiration, and improved antibiotics.
- In modern series, the mortality is typically $<15\%$.
- Significant sequelae, including seizures, persisting weakness, aphasia, or mental impairment, occur in 20% of survivors.

Thank you